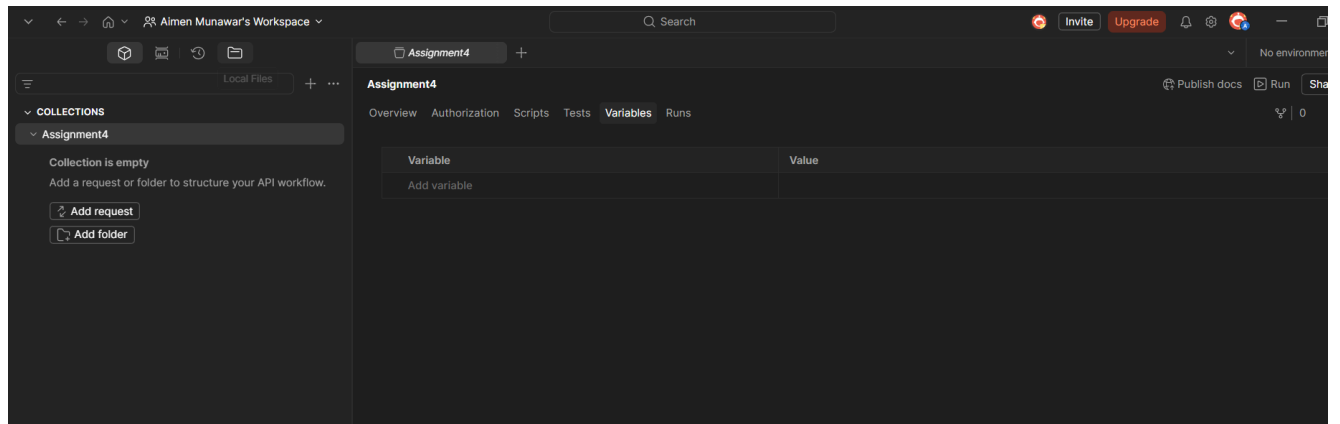


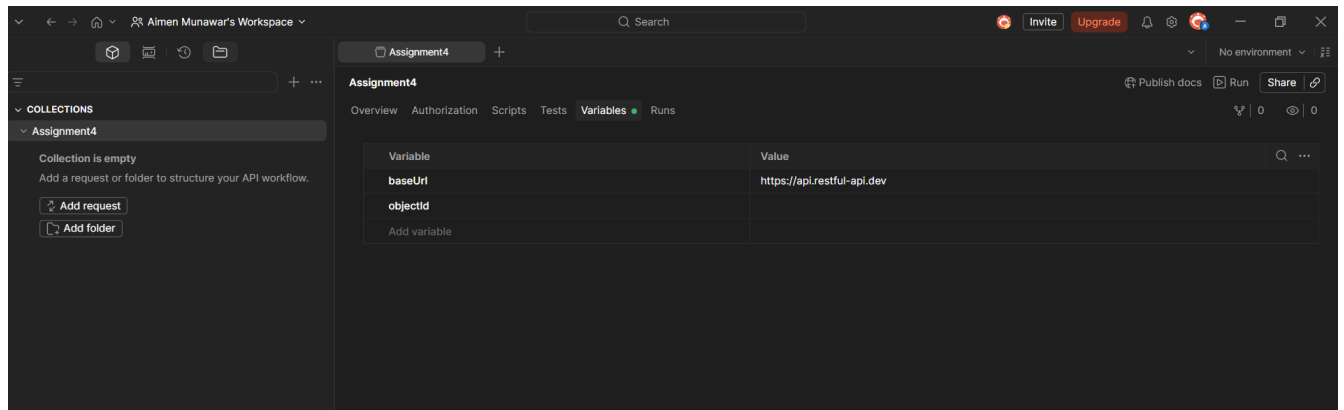
Assignment 4: API Testing with Postman

Name: Aimen Munawar

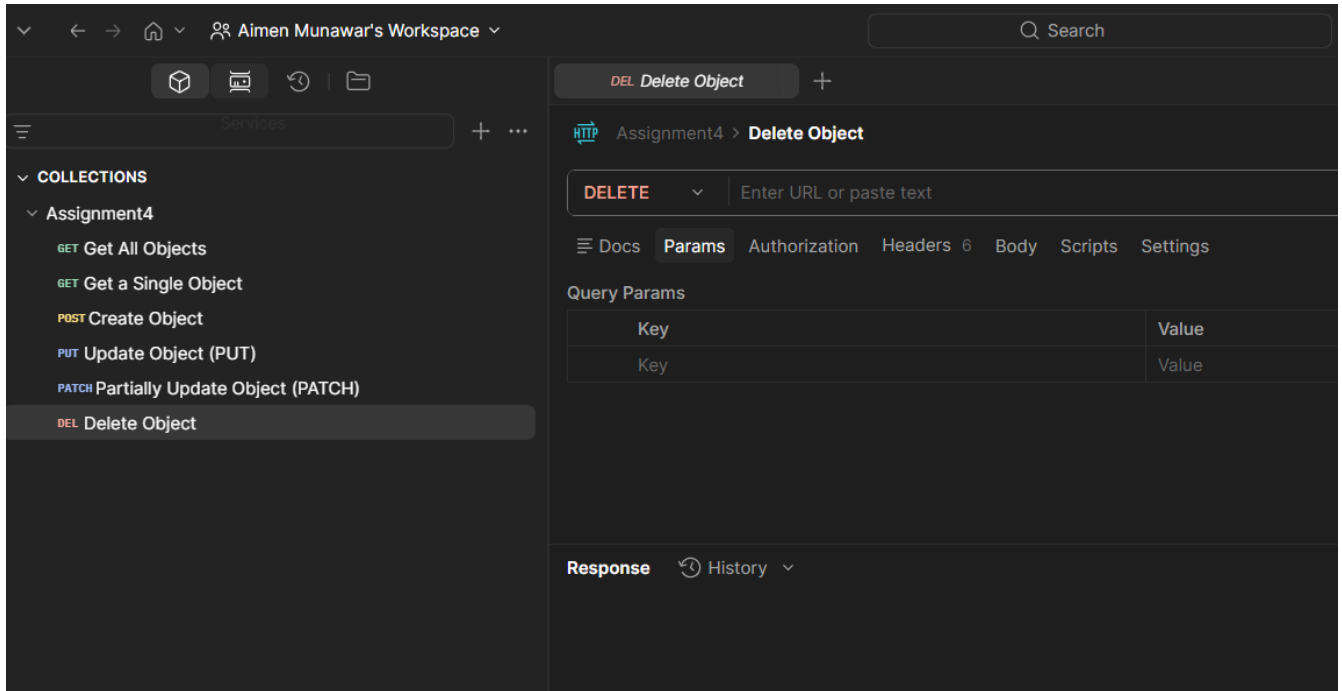
Creating Collection:



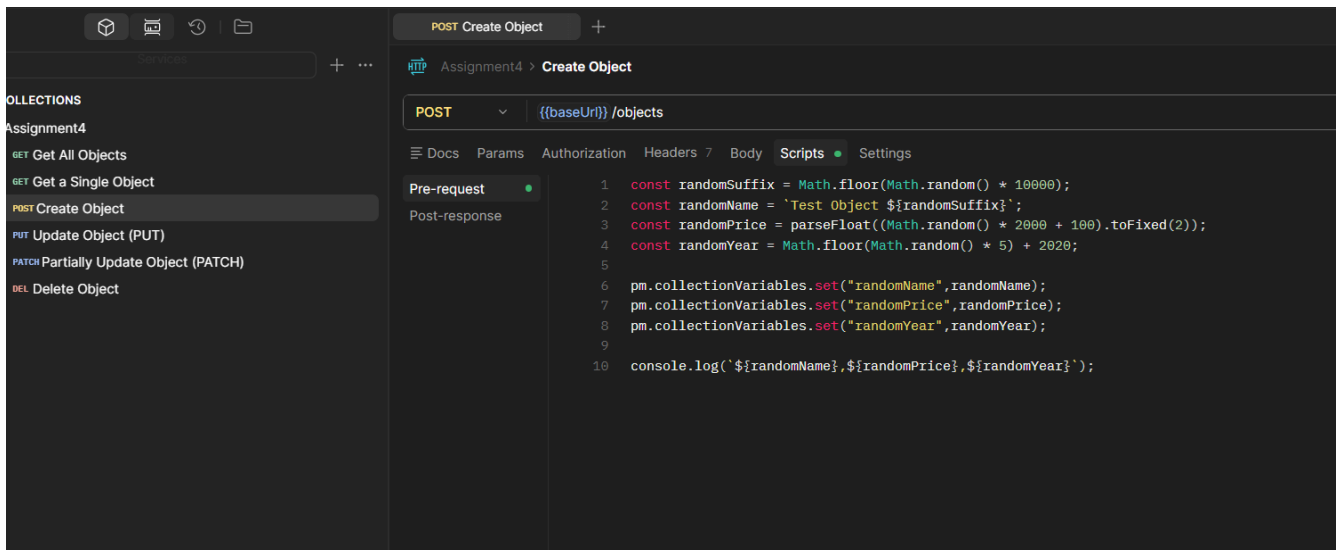
Adding Variables in Collection:

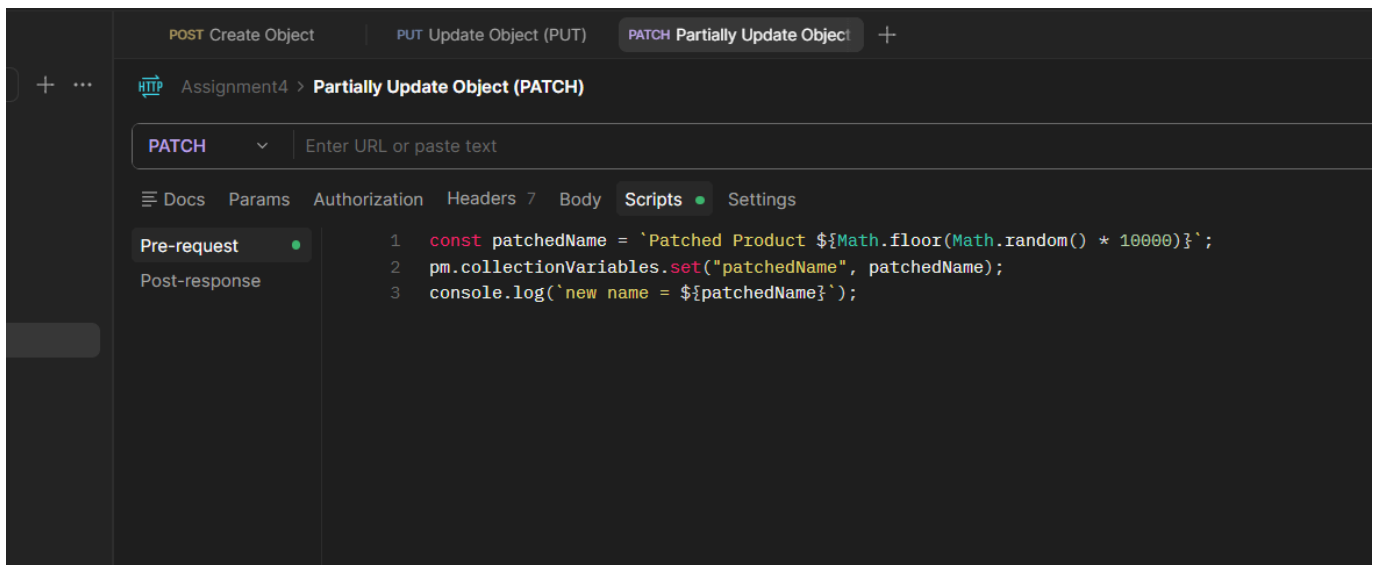
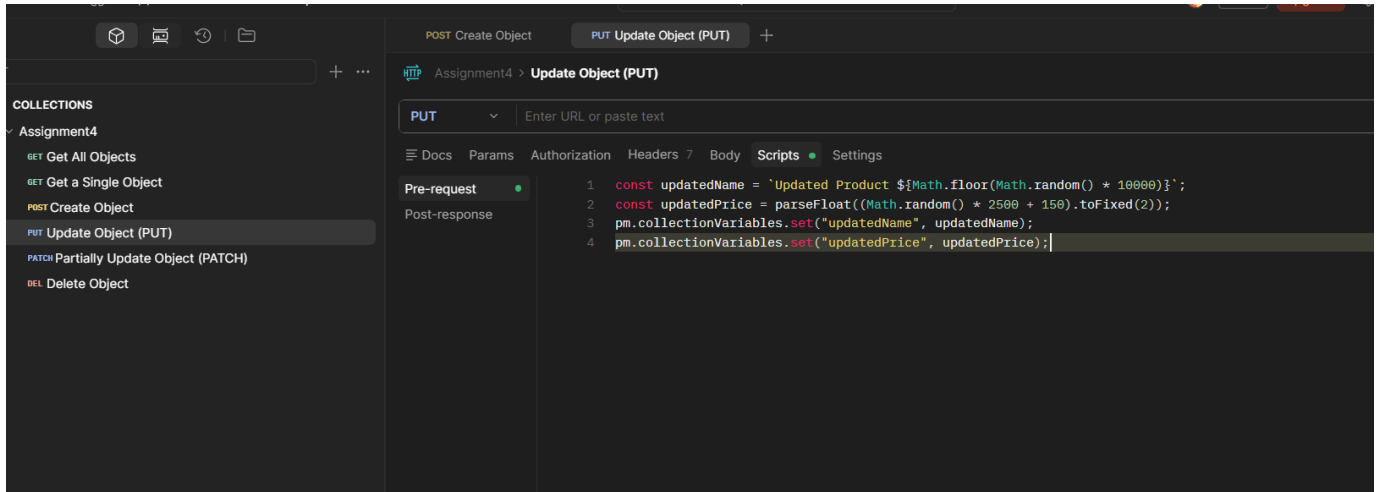


Building the Requests:

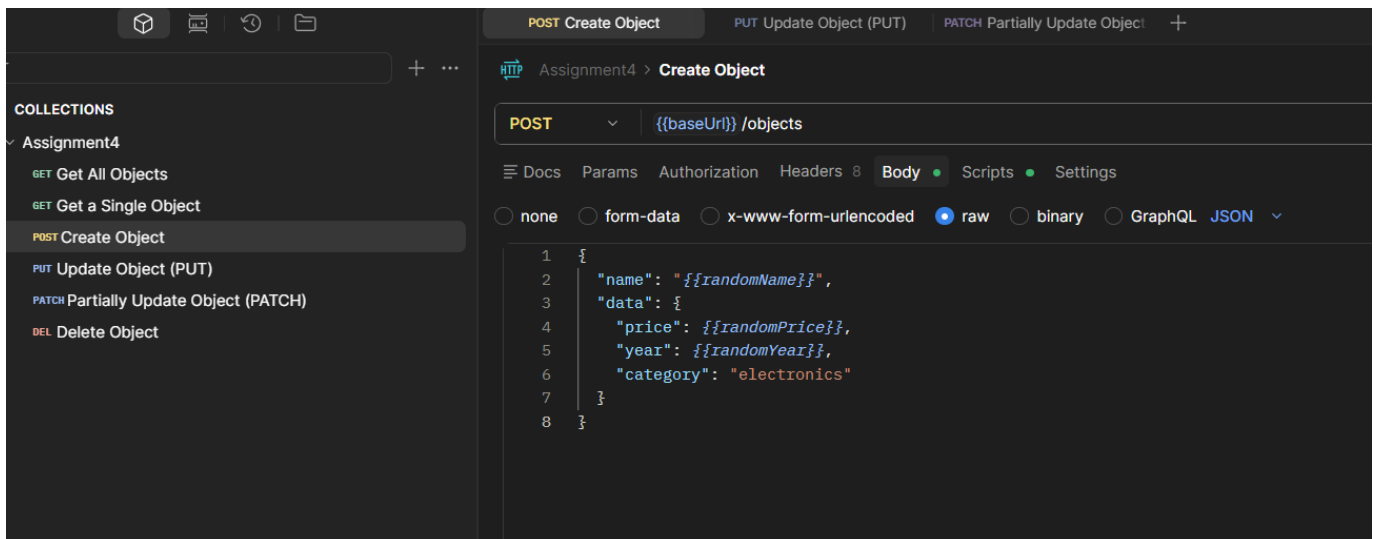


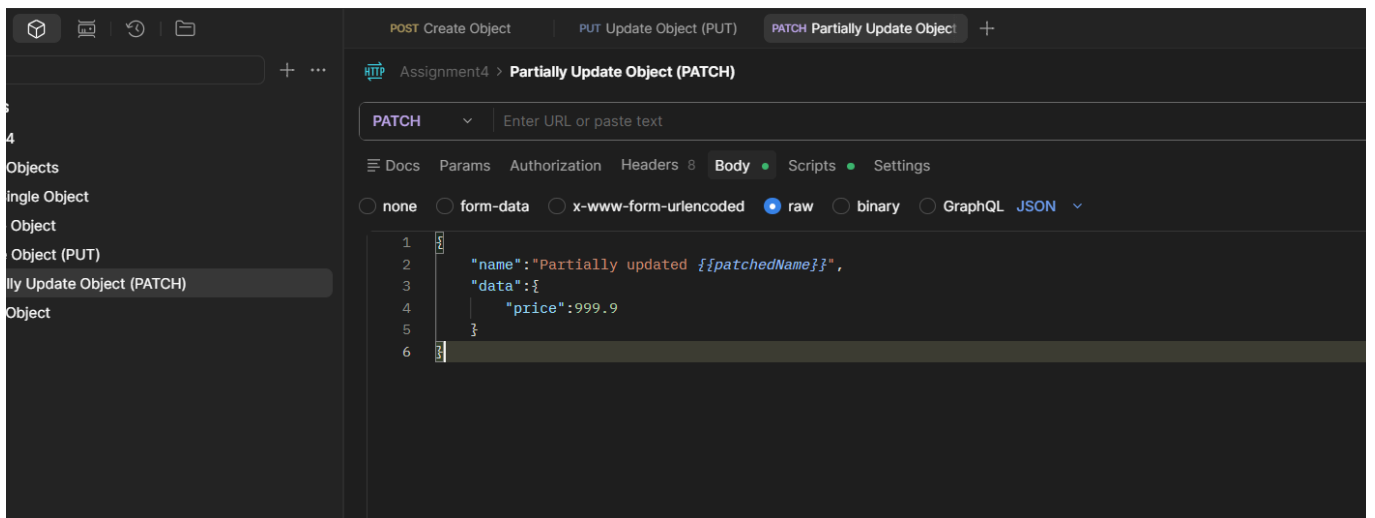
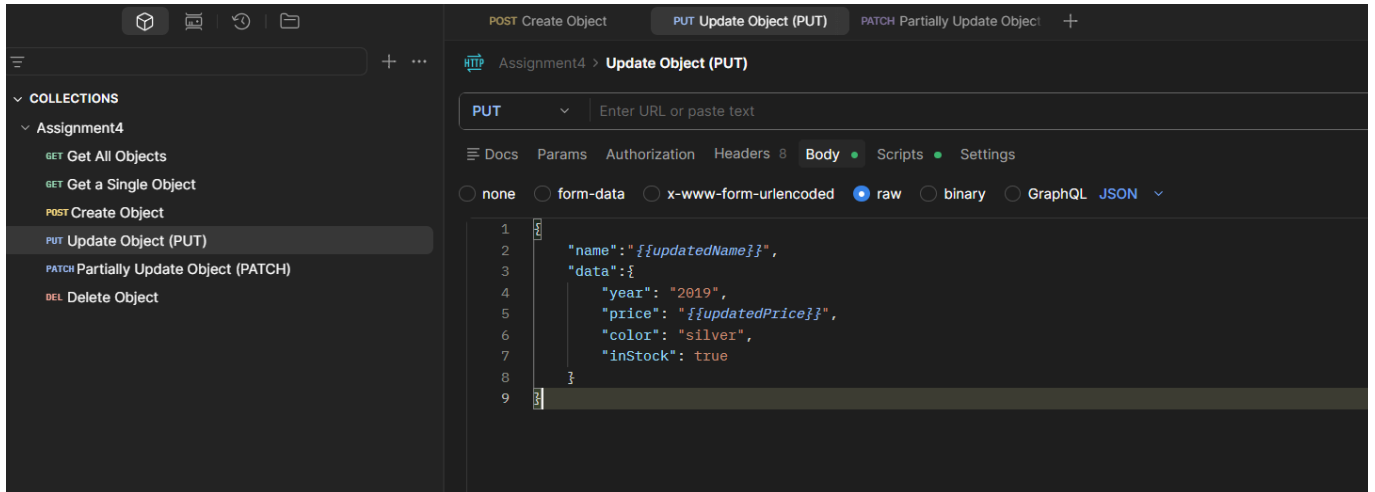
Adding Pre-request Scripts for Random Data:





Writing Request Bodies Using Variables:





Writing Tests with Chai Assertions, Logging & Chaining:

POST Create Object

PUT Update Object (PUT)

PATCH Partially Update Object

GET Get All Objects



Assignment4 > Create Object

POST



{{baseUrl}} /objects



Docs

Params

Authorization

Headers 8

Body

Scripts

Settings

Pre-request



Post-response



```
1  const response = pm.response.json();
2  console.log("Created Object");
3  console.log(response);
4
5  pm.test("Status code is 200 or 201", () => {
6    pm.response.to.have.status(200);
7  });
8
9  pm.test("Response contains id and name", () => {
10    pm.expect(response).to.have.property("id");
11    pm.expect(response).to.have.property("name");
12  });
13
14  pm.test("Generated name matches request", () => {
15    const expectedName = pm.collectionVariables.get("randomName");
16    pm.expect(response.name).to.equal(expectedName);
17  });
18
19  pm.test("Data.price matches generated value", () => {
20    const expectedPrice = pm.collectionVariables.get("randomPrice");
21    pm.expect(response.data.price).to.equal(expectedPrice);
22  });
23
24  pm.collectionVariables.set("objectId", response.id);
25  console.log(`Saved objectId = ${response.id}`);
26
27  pm.test("Failing test example - 1 should equal 2", () => {
28    pm.expect(1).to.equal(2);
```

Assignment4 GET Get a Single Object GET Get All Objects +

HTTP Assignment4 > Get All Objects

GET {{baseUrl}} /objects

Docs Params Authorization Headers 6 Body Scripts Settings

Pre-request
Post-response

```
1 const objects = pm.response.json();
2 console.log(`Total objects in the DB: ${objects.length}`);
3
4 pm.test("Status code is 200", () => {
5   pm.response.to.have.status(200);
6 });
7
8 pm.test("Response is an array", () => {
9   pm.expect(objects).to.be.an("array");
10 });
11
12 if (objects.length > 0) {
13   console.log("First object in list:", objects[0].name);
14   pm.test("First object has id and name", () => {
15     pm.expect(objects[0]).to.have.property("id");
16     pm.expect(objects[0]).to.have.property("name");
17   });
18 } else {
19   console.warn("No objects found - run Create Object first");
20 }
```

Assignment4 GET Get a Single Object GET Get All Objects +

HTTP Assignment4 > Get a Single Object

GET {{baseUrl}} /objects/ {{objectId}}

Docs Params Authorization Headers 6 Body Scripts Settings

Pre-request
Post-response

```
1 const obj = pm.response.json();
2 const expectedId = pm.collectionVariables.get("objectId");
3
4 console.log("Retrieved Single Object");
5 console.log(`ID: ${obj.id}, Name: ${obj.name}`);
6
7 pm.test("Status code is 200", () => {
8   pm.response.to.have.status(200);
9 });
10
11 pm.test("Retrieved ID matches chained objectId", () => {
12   pm.expect(obj.id).to.equal(expectedId);
13 });
14
15 pm.test("Object has name and data properties", () => {
16   pm.expect(obj).to.have.property("name");
17   pm.expect(obj).to.have.property("data");
18 });
19 console.log("Data:", obj.data);
```

Assignment4 PUT Update Object (PUT) PATCH Partially Update Object DEL Delete Object POST Create Object +

Assignment4 > Update Object (PUT)

PUT {{baseUrl}} /objects/ {{objectId}}

Docs Params Authorization Headers 8 Body Scripts Settings

Pre-request Post-response

```
1 const updated = pm.response.json();
2 const expectedId = pm.collectionVariables.get("objectId");
3
4 console.log("PUT Update Response");
5 console.log(updated);
6
7 pm.test("PUT returns status 200", () => {
8   pm.response.to.have.status(200);
9 });
10
11 pm.test("ID remains unchanged", () => {
12   pm.expect(updated.id).to.equal(expectedId);
13 });
14
15 pm.test("Name was updated to the random value", () => {
16   const newName = pm.collectionVariables.get("updatedName");
17   pm.expect(updated.name).to.equal(newName);
18 });
19
20 pm.test("Data.price was updated", () => {
21   const newPrice = parseFloat(pm.collectionVariables.get("updatedPrice"));
22   pm.expect(Number(updated.data.price)).to.equal(newPrice);
23 });
24
25 pm.test("New fields exist after PUT", () => {
26   pm.expect(updated.data).to.have.property("color", "silver");
27   pm.expect(updated.data).to.have.property("inStock", true);
28 });
```

Assignment4 PUT Update Object (PUT) PATCH Partially Update Object DEL Delete Object POST Create Object +

Assignment4 > Partially Update Object (PATCH)

PATCH {{baseUrl}} /objects/ {{objectId}}

Docs Params Authorization Headers 8 Body Scripts Settings

Pre-request Post-response

```
3
4 console.log("PATCH Partial Update Response");
5 console.log(patched);
6
7 pm.test("PATCH returns status 200", () => {
8   pm.response.to.have.status(200);
9 });
10
11 pm.test("ID did not change", () => {
12   pm.expect(patched.id).to.equal(expectedId);
13 });
14
15 pm.test("Name was partially updated", () => {
16   pm.expect(patched.name).to.include("Partially updated");
17 });
18
19 pm.test("Data.price changed to 999.9", () => {
20   pm.expect(patched.data.price).to.equal(999.9);
21 });
```

Assignment4
PUT Update Object (PUT)
PATCH Partially Update Object
DEL Delete Object
POST Create Object
+

HTTP
Assignment4 > Delete Object

DELETE
{{baseUrl}} /objects/ {{objectId}}

Docs
Params
Authorization
Headers 6
Body
Scripts
Settings

Pre-request
Post-response

```

1  const delResponse = pm.response.json();
2  const deletedId = pm.collectionVariables.get("objectId");
3
4  console.log("DELETE Response");
5  console.log(delResponse);
6
7  pm.test("DELETE returns status 200", () => {
8    pm.response.to.have.status(200);
9  });
10
11 pm.test("Confirmation message received", () => {
12   pm.expect(delResponse.message).to.include("deleted");
13 });
14 pm.sendRequest({
15   url: `${pm.collectionVariables.get("baseUrl")}/objects/${deletedId}`,
16   method: "GET"
17 }, (err, res) => {
18   pm.test("Object no longer exists - returns 404", () => {
19     pm.expect(res.code).to.equal(404);
20   });
21   console.log(`Verification GET status: ${res.code}`);
22 });

```

Running the Collection and Verify Chaining:

Assignment4
PUT Update Object (PUT)
PATCH Partially Update Object
DEL Delete Object
POST Create Object
+
No environment

Assignment4 - Run results
Run Again
+ New Run
Share

Ran today at 02:31:42 PM · View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	2s 160ms	23	0	191 ms

All 23
Passed 22
Failed 1
Skipped 0
Errors 0
Console log
List

Iteration 1

GET Get All Objects
https://api.restful-api.dev/objects
200 • 191 ms • 1.371 KB 3

PASS Status code is 200
PASS Response is an array
PASS First object has id and name

POST Create Object
https://api.restful-api.dev/objects
200 • 179 ms • 978 B 4 1

PASS Status code is 200 or 201
PASS Response contains id and name
PASS Generated name matches request
PASS Data.price matches generated value
FAIL Failing test example - 1 should equal 2 | AssertionError: expected 1 to equal 2

GET Get a Single Object
https://api.restful-api.dev/objects/ff8081819d82fab6019e06edbeb615ed
200 • 185 ms • 959 B 3

PASS Status code is 200
PASS Retrieved ID matches chained objectId
PASS Object has name and data properties

Assignment4

PUT Update Object (PUT)

PATCH Partially Update Object

DEL Delete Object

POST Create Object

+

No environment

Assignment4 - Run results

Ran today at 02:31:42 PM · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	2s 160ms	23	0	191 ms

All 23 **Passed 22** Failed 1 Skipped 0 Errors 0 Console log

List

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https://api.restful-api.dev/objects

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200 • 185 ms • 959 B 3

PASS Status code is 200

PASS Retrieved ID matches chained objectId

PASS Object has name and data properties

Assignment4

PUT Update Object (PUT)

PATCH Partially Update Object

DEL Delete Object

POST Create Object

Assignment4 - Run results

Ran today at 02:31:42 PM · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	2s 160ms	23	0	191 ms

All 23 **Passed 22** **Failed 1** Skipped 0 Errors 0 Console log

Iteration 1

POST Create Object

https://api.restful-api.dev/objects

FAIL Failing test example - 1 should equal 2 | AssertionError: expected 1 to equal 2

Assignment4

PUT Update Object (PU) PATCH Partially Update Object DEL Delete Object POST Create Object

No environment

Assignment4 - Run results

Ran today at 02:31:42 PM · View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	2s 160ms	23	0	191 ms

All 23 Passed 22 Failed 1 Skipped 0 Errors 0 Console log

Iteration 1

GET Get All Objects

https://api.restful-api.dev/objects

200 • 191 ms • 1.371 KB 3

LOG Total objects in the DB: 13

LOG First object in list: Google Pixel 6 Pro

POST Create Object

https://api.restful-api.dev/objects

200 • 179 ms • 978 B 4 1

LOG Test Object 5036,1533.44,2022

LOG Created Object

LOG {"id":"ff8081819d82fab6019e06edbeb615ed","name":"Test Object 5036","createdAt":1778232704695,"data":{"price":1533.44,"year":2022,"category":"electronics"}}

LOG Saved objectId = ff8081819d82fab6019e06edbeb615ed

GET Get a Single Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e06edbeb615ed

200 • 185 ms • 959 B 3

LOG Retrieved Single Object

LOG ID: ff8081819d82fab6019e06edbeb615ed, Name: Test Object 5036

LOG Data: {"price":1533.44,"year":2022,"category":"electronics"}

PUT Update Object (PUT)

https://api.restful-api.dev/objects/ff8081819d82fab6019e06edbeb615ed

200 • 206 ms • 981 B 5

LOG PUT Update Response

LOG {"id":"ff8081819d82fab6019e06edbeb615ed","name":"Updated Product 5227","updatedAt":1778232705225,"data":{"year":"2019","price":"710.14","color":"silver","inSto...

Assignment4

Create new request

Overview Authorization Scripts Variables Runs

Variable	Value
baseUrl	https://api.restful-api.dev
objectId	ff8081819d82fab6019e06edbeb615ed
randomName	Test Object 5036
randomPrice	1533.44
randomYear	2022
updatedName	Updated Product 5227
updatedPrice	710.14
patchedName	Patched Product 2086
Add variable	